

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question
 $H = 120p + 60$

The Karvonen formula above shows the relationship between Alice’s target heart rate H , in beats per minute (bpm), and the intensity level p of different activities. When $p = 0$, Alice has a resting heart rate. When $p = 1$, Alice has her maximum heart rate. It is recommended that p be between 0.5 and 0.85 for Alice when she trains. Which of the following inequalities describes Alice’s target training heart rate?

- Answer**
- A. $120 \leq H \leq 162$
 - B. $102 \leq H \leq 120$
 - C. $60 \leq H \leq 162$
 - D. $60 \leq H \leq 102$

Correct Answer: A

Rationale

Choice A is correct. When Alice trains, it’s recommended that p be between 0.5 and 0.85. Therefore, her target training heart rate is represented by the values of H corresponding to $0.5 \leq p \leq 0.85$. When $p = 0.5$, $H = 120(0.5) + 60$, or $H = 120$. When $p = 0.85$, $H = 120(0.85) + 60$, or $H = 162$. Therefore, the inequality that describes Alice’s target training heart rate is $120 \leq H \leq 162$.

Choice B is incorrect. This inequality describes Alice’s target heart rate for $0.35 \leq p \leq 0.5$. Choice C is incorrect. This inequality describes her target heart rate for $0 \leq p \leq 0.85$. Choice D is incorrect. This inequality describes her target heart rate for $0 \leq p \leq 0.35$.